LLM Performance Analysis

on Real-World Legal Scenario Interpretation

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1. Objective

The goal of this project is to compare multiple Large Language Models (LLMs) on the same real-world task and evaluate differences in quality, usefulness, reasoning, hallucination risk, and practical applicability. The evaluation focuses on the following critical dimensions:

- Legal accuracy and correctness of statutory references
- Depth and structure of legal reasoning
- Clarity of writing and response organisation
- Adherence to prompt instructions
- Ability to handle ambiguity without overconfidence
- Risk of hallucination or fabricated citations
- Practical usefulness in a real-world legal context

This project moves beyond superficial comparisons to assess how effectively each model simulates judicial reasoning and structured legal analysis.

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2. Chosen Real-World Task

Selected Task

Legal analysis of a criminal self-defense case involving a minor.

Why This Task Was Chosen

This task demands a combination of legal knowledge, logical reasoning, ethical evaluation, and the ability to reach conclusions from incomplete facts — making it an ideal stress-test for LLM reasoning. Specific reasons include:

- High-stakes, consequential decision-making (criminal law)
- Requires structured judicial thinking rather than general knowledge retrieval
- Tests whether models can avoid emotional bias and unwarranted assumptions
- Involves ambiguity — an ideal probe for intellectual honesty

Real-World Applicability

Such analysis is directly applicable in the following contexts:

- Legal research assistance and academic education
- Case preparation and legal drafting support
- AI-assisted judicial decision-support systems
- Preliminary legal advisory tools for the public

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3. Models Selected

The four models below were chosen to represent a range of providers, pricing tiers, and architectural approaches — enabling a fair and diverse comparison.

Model Name	Provider	Access Type	Plan
ChatGPT (GPT-5.3 Go)	OpenAI	Paid	Go Plan
Gemini 3.1 Pro	Google	Paid	AI Pro Plan
Claude Sonnet 4.6	Anthropic	Free	Free Plan
Perplexity Sonar	Perplexity AI	Free	Free Plan

4. Prompt Used

The identical prompt below was submitted to all four models without any modification, ensuring a fair and controlled comparison.

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You are acting as a judge evaluating a criminal case involving self-defense.

Case Details:
A minor girl was alone at home while her parents were away. A male relative
visited the
house, initially behaving normally and asking for coffee. After some time, he
attempted
to sexually assault (rape) her. During the struggle, the girl defended
herself by hitting
him with a rod. The attacker died due to the injury.

Your task is to analyze this case from a legal and ethical perspective.

Instructions:
  Identify the key legal issue(s) involved
  Evaluate whether this qualifies as self-defense (mention legal principles,
    e.g., right of private defense)
  Analyze proportionality of force used
  Consider the role of the victim being a minor
  Provide a reasoned judgment (not just opinion)
  Clearly state whether this is likely to be considered a crime or justified
action
  Highlight any uncertainties or assumptions in the case

Important:
  Do not give a purely emotional answer
  Do not assume facts not given
  Avoid overconfidence; acknowledge ambiguity where present
  Focus on structured reasoning and real-world applicability
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5. Benchmark Check

Background research was conducted on each model prior to the task evaluation to calibrate expectations against actual performance.

Model 1 — ChatGPT (GPT-5.3 Go)

Attribute	Details
Provider	OpenAI
Context Window	Large
Speed / Latency	Moderate to Fast
Price	Paid (Go Plan)
General Positioning	Strong reasoning, structured output
Benchmark Observation	High performance in reasoning tasks and instruction-following
Real Task Observation	Clear, structured, legally accurate, well-balanced conclusions

Model 2 — Gemini 3.1 Pro

Attribute	Details
Provider	Google
Context Window	Very Large
Speed / Latency	Fast
Price	Paid (AI Pro Plan)
General Positioning	General-purpose, fast, multimodal
Benchmark Observation	Strong general performance across diverse tasks
Real Task Observation	Good legal clarity but shallower reasoning depth

Model 3 — Claude Sonnet 4.6

Attribute	Details
Provider	Anthropic
Context Window	Very Large
Speed / Latency	Moderate
Price	Free Plan
General Positioning	Deep analytical reasoning, low hallucination rate
Benchmark Observation	Strong reasoning and instruction adherence
Real Task Observation	Deepest and most nuanced legal analysis of all four models

Model 4 — Perplexity Sonar

Attribute	Details
Provider	Perplexity AI
Context Window	Moderate
Speed / Latency	Fast
Price	Free Plan
General Positioning	Search-augmented AI, strong factual retrieval
Benchmark Observation	Excellent at citing sources; weaker at synthesis
Real Task Observation	Informational but lacks depth of reasoning for legal analysis

6. Output Comparison

Below are the detailed responses from each model to the identical prompt, preserved as faithfully as possible to the original structure and content.

Model 1 Output — ChatGPT (GPT-5.3 Go)

> 1. KEY LEGAL ISSUES
> 1. Right of Private Defense — Whether the girl's act falls under IPC Sections 96–106.
> 2. Extent of Force Used — Whether striking with a rod causing death was necessary and proportionate.
> 3. Imminence of Threat — Whether there was reasonable apprehension of rape.
> 4. Culpability for Death — Whether the resulting death constitutes no offense, culpable homicide, or exceeds the scope of private defense.
>
> 2. DOES THIS QUALIFY AS SELF-DEFENSE?
> The right of private defense arises when there is a reasonable apprehension of imminent harm.
> IPC Section 100 explicitly recognizes that defense against rape permits even causing the death of the attacker. The threat was immediate and real. The girl acted during the assault, not after.
> Conclusion: This strongly supports that her actions fall within the right of private defense.
>
> 3. PROPORTIONALITY OF FORCE
> In cases involving rape or attempted rape, the law allows lethal force if necessary.
> In a physical struggle, a victim may not have the luxury of calibrated force.
> A minor facing sexual assault is in a high-risk, panic-driven situation.
> Conclusion: The use of force is likely legally proportionate given the severity of the threat.
>
> 4. ROLE OF THE VICTIM BEING A MINOR
> A minor is more vulnerable and less capable of measured response.
> Courts assess actions from the perspective of a reasonable person in the victim's position, including age and psychological capacity. Her minor status strengthens the justification.
>
> 5. REASONED JUDGMENT
> The act is likely legally justified under the right of private defense — not a criminal offense — provided force was used during the assault with no evidence of excessive retaliation.

6. CRIME OR JUSTIFIED ACTION?
Most likely classification: Justified action (no offense) under right of private defense.
Could be questioned if the assault had already ceased before the fatal blow, or if force was clearly excessive (e.g., repeated blows after incapacitation).

7. UNCERTAINTIES
— Was it clearly an attempt to rape or ambiguous conduct?
— Did the fatal blow occur during the struggle or after?
— Single defensive strike vs. repeated attacks
— Cause of death and medical evidence
— Whether retreat was realistically possible

FINAL POSITION: The girl's actions are highly likely protected under right of private defense. Most likely not a crime — a legally justified act of self-defense, subject to factual verification.

Model 2 Output — Claude Sonnet 4.6

Case Analysis: Minor Girl vs. Deceased Relative — Self-Defense in Sexual Assault Context

1. KEY LEGAL ISSUES
Primary: Does the girl's act of striking the attacker with a rod, resulting in his death, constitute a criminal offense, or is it protected under the right of private defense?
Secondary: imminence of threat, minor's judgment and capacity, proportionality of force, and whether the attacker's death negates the defense or is incidental to it.

2. RIGHT OF PRIVATE DEFENSE — LEGAL FRAMEWORK (IPC SECTIONS 96–106)
Reasonable apprehension of harm → Attempted sexual assault: severe threat to bodily integrity
No time to seek police protection → She was alone at home — no avenue for immediate help
Right exists even if attacker dies → IPC S.100: fatal force permitted in rape/kidnapping cases

Under Section 100 IPC, the right extends to causing death when there is reasonable apprehension of grievous hurt, assault with intent to commit rape, kidnapping, wrongful confinement, or acid attack. An attempted rape falls squarely within this provision.

3. PROPORTIONALITY OF FORCE
Supporting proportionality:
— Threat was not merely verbal; physical assault had already commenced
— Girl was alone, likely smaller and weaker than the adult male attacker
— A rod is a household object — not a premeditated weapon
— Intent was survival, not retribution — she struck during active struggle
Against proportionality (counter-arguments that may be raised):
— Death resulted — courts will examine carefully
— Number of strikes matters; post-incapacitation blows would be problematic
— Indian law does NOT impose a duty to retreat inside one's own home
Assessment: Proportionality is likely satisfied — force was used during the assault, not after.

4. ROLE OF BEING A MINOR
Legally: More lenient standard of reasonableness applies; POCSO Act (2012) affords special

protections to child victims; a minor cannot consent to sexual acts under any circumstances.
Ethically: Extreme power imbalance between an adult male relative and a lone minor at home.

5. REASONED JUDGMENT (5 Findings)
Finding 1 — Imminent and severe threat: Attempted rape by physically stronger adult.
Finding 2 — Legal coverage under IPC S.100: Not a gray area in the statute.
Finding 3 — No reasonable alternative: Alone, no access to help, already in physical struggle.
Finding 4 — Minor status: Reduces standard expected; invokes POCSO protective intent.
Finding 5 — Absence of malice: No premeditation; no post-assault continuation of violence.
Probable outcome: Court would most likely acquit or decline to charge.

6. UNCERTAINTIES
— How many times she struck him (multiple post-subdual strikes raises disproportionality)
— Whether he was incapacitated when the fatal blow landed
— Presence of witnesses or forensic corroboration
— Exact age of minor (affects juvenile vs. adult court proceedings)

CONCLUSION: This is not a crime. It is a justified act of private defense. The appropriate judicial response is acquittal — the burden of guilt lies entirely on the deceased for initiating the assault.

Model 3 Output — Gemini 3.1 Pro

As the presiding judge, I must balance the gravity of a lost life against the fundamental right of an individual to protect their bodily integrity.

1. KEY LEGAL ISSUES
Actus Reus: The girl caused the death of the relative (Homicide).
Justification: Was the homicide legally justified under the right of private defense?
Mens Rea: Did the girl intend to kill, or was her intention to repel the sexual assault?

2. EVALUATION OF PRIVATE DEFENSE (IPC SECTIONS 96–106)
The right extends to causing death if the offense involves an assault with intention of rape.
Apprehension of Danger: The law does not require the victim to wait for the assault to complete.
The Relative's Status: Breach of trust reinforces the victim's vulnerability in a domestic setting.

3. PROPORTIONALITY OF FORCE
The 'Golden Scales' Principle: A person facing a traumatising assault cannot be expected to modulate their defense with mathematical precision.
The Instrument: A rod is a weapon of opportunity — a reflexive and proportionate response.
The Outcome: If the blow was struck to stop the attack, resulting death is an unfortunate consequence of a justified defense, not an act of murder.

4. STATUS AS A MINOR
Physical Disparity: Presumed power imbalance makes the use of a weapon more justifiable.
Psychological Impact: A minor is more likely to experience reasonable apprehension of extreme harm during a violent struggle, validating a more forceful defensive reaction.

5. REASONED JUDGMENT — Verdict: Justified Action (Right of Private Defense)
1. Immediacy: The threat to her safety was immediate and grave.

2. No Retreat Required: One is not required to retreat from an attacker in their own home (Castle Doctrine applies).
3. Lack of Malice: No evidence of premeditation; the defense was a spontaneous reaction.

UNCERTAINTIES
— Number of blows: Continued striking after incapacitation could shift to punitive action.
— Medical cause of death: Assumed rod strike was direct cause.
— Means of escape: Assumed no other reasonable exit was available.

CONCLUSION: This case is unlikely to be considered a crime. It is a classic exercise of the Right of Private Defense. The law does not require a victim to submit to rape; she is entitled to use all necessary force to protect her personhood.

Model 4 Output — Perplexity Sonar

This case involves a clear act of self-defense under Indian law against an attempted sexual assault. The right of private defense applies here, as the victim was a minor girl alone at home facing an imminent threat from a male relative.

KEY LEGAL ISSUES
The primary issue is whether the girl's actions fall under the right of private defense (Sections 96–106 IPC, now BNS Sections 34–44), specifically excusing homicide in cases of assault intended to commit rape (IPC Section 100, thirdly).
No duty to retreat exists when immediate harm is apprehended, but force must remain defensive.
The case hinges on the timing of the struggle and the necessity of the rod as a weapon.

SELF-DEFENSE EVALUATION
Indian law permits causing death in private defense if there is reasonable apprehension of death, grievous hurt, or specifically rape (IPC Section 99).
The attempted rape creates such apprehension — as seen in State of Orissa v. Nirupama Panda.
Courts assess good faith and immediacy based on circumstances, not requiring exact proportionality.

PROPORTIONALITY ANALYSIS
Against a rape threat, deadly force is explicitly allowed under IPC Section 100.
Hitting with a rod during a struggle aligns with necessity.
Disproportionate force = pursuit or excess blows post-threat cessation — not indicated here.

MINOR VICTIM'S ROLE
The right of private defense extends equally to minors; age does not limit it.
Her minor status heightens the reasonableness of fearing grievous harm.
No special culpability attaches to minors acting in self-preservation during such assaults.

REASONED JUDGMENT
This qualifies as justified self-defense, not a crime — the act meets IPC Section 100 criteria.
The death resulted from proportionate defensive force in a sudden struggle.
Likely acquittal in court, mirroring precedents, absent evidence of malice.

UNCERTAINTIES
— Exact injury details (one blow vs. multiple) and precise struggle sequence

> — Forensic proof of ongoing assault at the moment of death
> — Assumed 'minor' means under 18; relative's intent proven only by given facts

7. Analysis

Comparative Summary Table

Criteria	ChatGPT	Claude	Gemini	Perplexity
Legal Accuracy	Strong	Very Strong	Good	Mixed
Reasoning Depth	High	Very High	Moderate	Low
Clarity	Excellent	Verbose	Clear	Fragmented
Prompt Adherence	Strong	Very Strong	Good	Partial
Practical Usefulness	High	Very High	Moderate	Low
Ambiguity Handling	Good	Excellent	Limited	Weak
Hallucination Risk	Low	Very Low	Moderate	High

Key Questions — Direct Answers

Clearest Response: ChatGPT — well-structured, concise, easy to follow
Best Prompt Adherence: Claude — addressed every instruction point methodically
Most Useful for Real-World Use: Claude — deepest legal analysis with actionable findings
Hallucination / Overconfidence: Perplexity (highest risk); Claude (minor overconfidence)
Best Ambiguity Handling: Claude — explicitly listed all uncertainties and legal impact

Key Strengths Per Model

- ChatGPT: Best overall balance of clarity, structure, and legal accuracy
- Claude: Deepest reasoning, lowest hallucination risk, best ambiguity handling
- Gemini: Good general coverage; fast and readable output
- Perplexity: Strong at retrieving and citing factual legal references

Key Weaknesses Per Model

- ChatGPT: Slightly shallower than Claude for nuanced multi-factor legal analysis
- Claude: Verbose — denser to read; occasional slight overconfidence in verdict tone
- Gemini: Lacks depth for complex, layered reasoning tasks
- Perplexity: Unsuitable for reasoning-heavy tasks; over-relies on citations over analysis

8. Final Verdict

Verdict Category	Assessment
Best Performing Model	Claude Sonnet 4.6 — deepest reasoning, best prompt adherence, highest practical utility
Runner-Up	ChatGPT (GPT-5.3 Go) — clearest structure, strongest overall clarity
Adequate Performance	Gemini 3.1 Pro — acceptable general analysis; limited in depth
Weakest Performing Model	Perplexity Sonar — insufficient for legal reasoning tasks
Benchmark vs Real Output	Benchmark expectations largely matched real-world results across all four models

Final Recommendation

Based on this evaluation, the following recommendations are made for practitioners choosing an LLM for legal analysis tasks:

- **Use Claude — for high-stakes legal reasoning, ambiguity handling, and detailed judicial analysis**
- Use ChatGPT — for clarity-first tasks, drafting, and well-structured legal summaries
- Use Gemini — for general-purpose tasks where speed and multimodal capability matter more than depth
- Avoid Perplexity — for any task requiring structured reasoning, legal interpretation, or analytical depth

Claude demonstrated that free-tier access does not imply lower quality — it outperformed both paid models on the dimensions that matter most for legal reasoning.